

Learning Koopman Operators

①

Given an autonomous nonlinear dynamical sys:

$$\boxed{\begin{array}{l} x_0 \in \Omega \subset \mathbb{R}^d \\ F: \Omega \rightarrow \Omega \end{array}}$$

$$x_{k+1} = F(x_k), \quad k=0,1,2,\dots,$$

Suppose that we have snapshot data

$$Y = \begin{bmatrix} 1 & \dots & 1 \\ y_1 & \dots & y_m \\ 1 & \dots & 1 \end{bmatrix}, \quad X = \begin{bmatrix} 1 & \dots & 1 \\ x_1 & \dots & x_m \\ 1 & \dots & 1 \end{bmatrix} ~~\in \mathbb{R}^{(m+1) \times d}~~$$

where $y_k = F(x_k)$ for $k=1,\dots,m$.

Q: Can we build an approximation to the Koopman operator?

Idea: Learn matrix representing K on

$$V_N = \text{span} \{ \varphi_1, \dots, \varphi_N \} \subset L^2(\Omega).$$

$$\Rightarrow L^2(\Omega) = \text{inner product space w/ } \langle f, g \rangle = \int_{\Omega} f \bar{g} dx$$

$$f \in L^2(\Omega) \Leftrightarrow \|f\|^2 = \langle f, f \rangle < \infty$$

$\Rightarrow$ Inner product space allows us to build finite-dim approx. via orthogonal proj.

Galerkin Approximation

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To "learn" $K_N: V_N \rightarrow V_N$ representing

the Koopman operator $K: L^2 \rightarrow L^2$ on V_N ,

we look at the action of K on each $\{\varphi_i\}_{i=1}^N$:

$$(i=1, \dots, N) \quad [K \varphi_i](x) = \varphi_i(F(x)) \quad \leftarrow [\in V_N, \text{ in general}]$$

$$= \sum_{i=1}^N [K_N]_{ij} \varphi_j(x) + \rho_j(x)$$

Idea: Choose j th column of K_N to minimize the residual ρ_j in the $L^2(\Omega)$ norm.

$$\left\| \begin{bmatrix} \varphi_1^F \\ \vdots \\ \varphi_N^F \end{bmatrix} - \begin{bmatrix} \varphi_1 \\ \vdots \\ \varphi_N \end{bmatrix} K_N \right\|_2 = \|\rho_j\| = \left\| \varphi_j(F(x)) - \sum_{i=1}^N [K_N]_{ij} \varphi_i(x) \right\|$$

quasimatrix notation

This is a least-squares problem in $L^2(\Omega)$ and the minimizer satisfies the orth. res. crit.

$$(\rho_j \perp V_N) \quad 0 = \langle \varphi_j^F, \varphi_i \rangle - \sum_{i=1}^N \langle \varphi_i, \varphi_i \rangle [K_N]_{ij}$$

$$\Phi^F - \Phi K_N \Rightarrow \Phi^* \Phi^F = \Phi^* \Phi K_N$$

Data-Driven Approximation

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In practice, we do not have observe $\phi_j(x)$ and $[K\phi_j](x) = \phi_j(f(x))$ at every point $x \in \Omega$, and we cannot compute $\langle \phi_i, \phi_j \rangle$ and $\langle \phi_i \circ f, \phi_j \rangle$ directly. Instead, we observe the pairs

$$\phi_j(\underline{x}_k) \text{ and } \phi_j(\underline{y}_k), \quad k=1, \dots, m.$$

Q: How can we approximate the required inner products $\langle \phi_i, \phi_j \rangle$ and $\langle \phi_i \circ f, \phi_j \rangle$ from the snapshot data X, Y ?

Idea: Approximate L^2 inner products w/ quadrature

$$\begin{aligned} \langle f, g \rangle &= \int_{\Omega} f(x) \overline{g(x)} dx \\ &\approx \sum_{k=1}^m w_k f(\underline{x}_k) \overline{g(\underline{x}_k)} \end{aligned}$$

Here, $\{w_k\}_{k=1}^m = \text{quad. weights}$

$\{\underline{x}_k\}_{k=1}^m = \text{quad. nodes (= input snap. data)}$

Weighted Least-Squares

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Applying our quadrature rule, we get

$$\langle \phi_i, \phi_j \rangle \approx \sum_{k=1}^m w_k \phi_i(x_k) \overline{\phi_j(x_k)}$$

$$\begin{aligned} \langle \phi_i \circ F, \phi_j \rangle &\approx \sum_{k=1}^m w_k \phi_i(F(x_k)) \overline{\phi_j(x_k)} \\ &= \sum_{k=1}^m w_k \phi_i(y_k) \overline{\phi_j(x_k)} \end{aligned}$$

For practical computation; analysis, it is useful to collect our results in matrix form.

$$\Phi_x = \begin{bmatrix} \phi_1(x_1) & \dots & \phi_N(x_1) \\ \vdots & & \vdots \\ \phi_1(x_m) & \dots & \phi_N(x_m) \end{bmatrix}, \quad \Phi_y = \begin{bmatrix} \phi_1(y_1) & \dots & \phi_N(y_1) \\ \vdots & & \vdots \\ \phi_1(y_m) & \dots & \phi_N(y_m) \end{bmatrix}$$

Let $W = \text{diag}(w_1, \dots, w_m)$, and calculate

$$\langle \phi_i, \phi_j \rangle = [\Phi_x^* W \Phi_x]_{ji} = G$$

$$\langle \phi_i \circ F, \phi_j \rangle = [\Phi_x^* W \Phi_y]_{ji} = A$$

"Weighted
Least-squares"
 $\Rightarrow$

$$G \hat{K}_N = A$$

"Data-Driven" Residual

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This weighted least-squares problem implies

$$\Phi_x^* W (\Phi_x \hat{K}_N - \Phi_Y) = 0,$$

which is equivalent to minimizing

$$\begin{aligned} \|W^{\frac{1}{2}} R\|_F^2 &= \sum_{i=1}^N \sum_{k=1}^m w_k |\rho_j(x_k)|^2 \\ &= \sum_{i=1}^N w_i \sum_{k=1}^m w_k \left| \varphi_j(x_k) - \sum_{i=1}^N [K_N^*]_{ij} \varphi_i(x_k) \right|^2 \end{aligned}$$

I.e., minimizing the residual on the data with respect to the weighted Frob. norm.